Week 9

Housing Data Cleaning & Transformation

Introduction :

Data cleaning and transformation are important steps in data analysis. In this project, the King County House Sales Dataset is cleaned and transformed using Power BI and Power Query Editor. The dataset contains information about house prices, sale dates, bedrooms, bathrooms, and property areas. New columns are created to organize the data and prepare it for further analysis.

Objectives :

- To import the housing dataset into Power BI.
- To clean missing and invalid data.
- To remove duplicate records.
- To extract Sale Year and Sale Month.
- To calculate Property Age and Price per SqFt.
- To calculate Total Area and identify renovated properties.
- To categorize properties by size and price.
- To prepare the data for further analysis.

Software Used :

- Power BI Desktop
- Power Query Editor

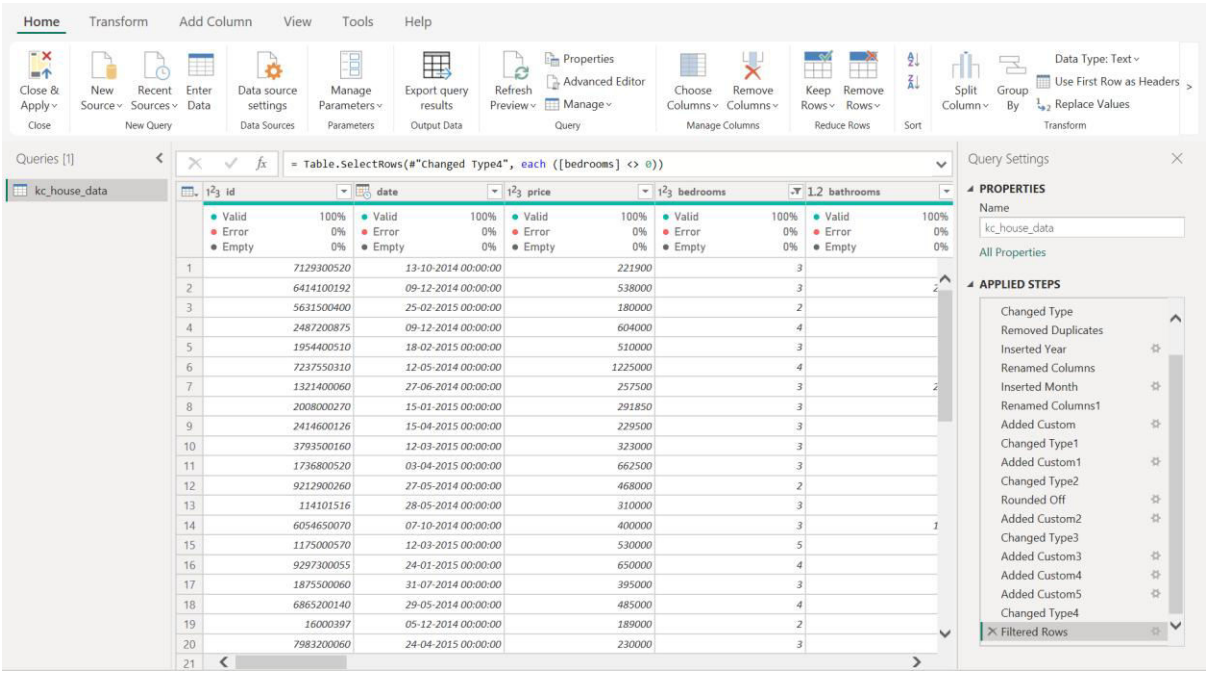
Dataset Description :

The King County House Sales Dataset contains historical house sales information. It includes property ID, sale date, price, bedrooms, bathrooms, living area, lot area, year built, and renovation year. This dataset helps analyze property characteristics and house prices.

Data Cleaning :

The dataset was imported into Power Query Editor for cleaning and transformation.

- Missing values were checked.
- Duplicate IDs were removed.
- Column data types were checked and adjusted where required.
- The sale date column was prepared for extracting date information.
- Records with zero bedrooms and zero bathrooms were filtered out.



Data Transformation :

➢ Sale Year :

The year was extracted from the sale date using the Date options in Power Query Editor. This column helps analyze house sales by year.

➢ Sale Month

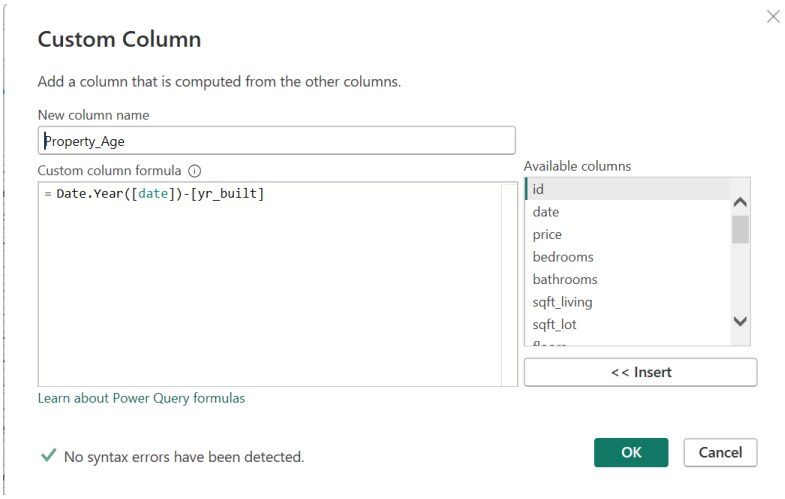
The month was extracted from the sale date. This helps organize and analyze house sales by month.

➢ Property Age

A custom column named Property_Age was created to calculate the age of the property at the time of sale.

Formula:

Date.Year([date]) - [yr_built]



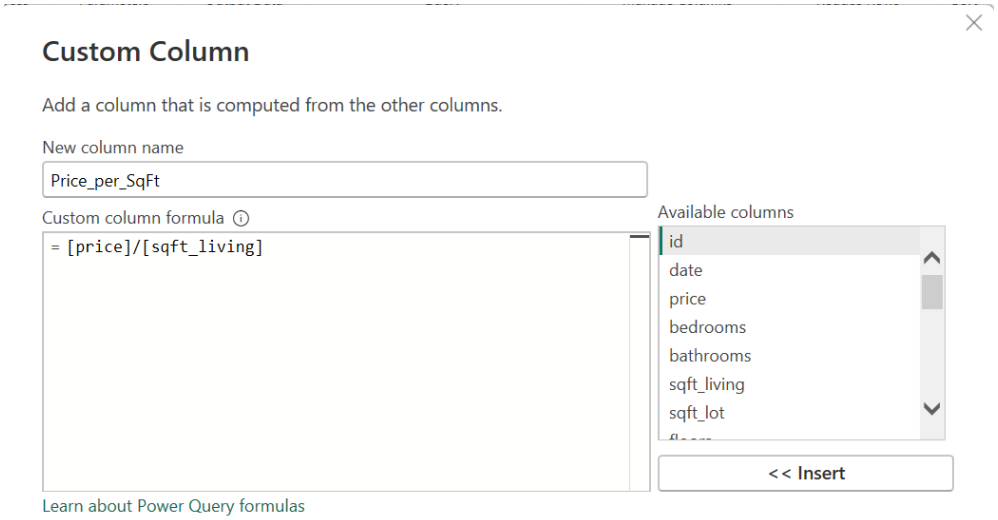
➢ Price per SqFt :

A custom column named Price_per_SqFt was created by dividing the house price by the living area.

Formula:

[price] / [sqft_living]

The result was rounded to two decimal places to make the values easier to read.



➢ Total Area :

A custom column named Total_Area was created by adding the living area and lot area.

Formula:

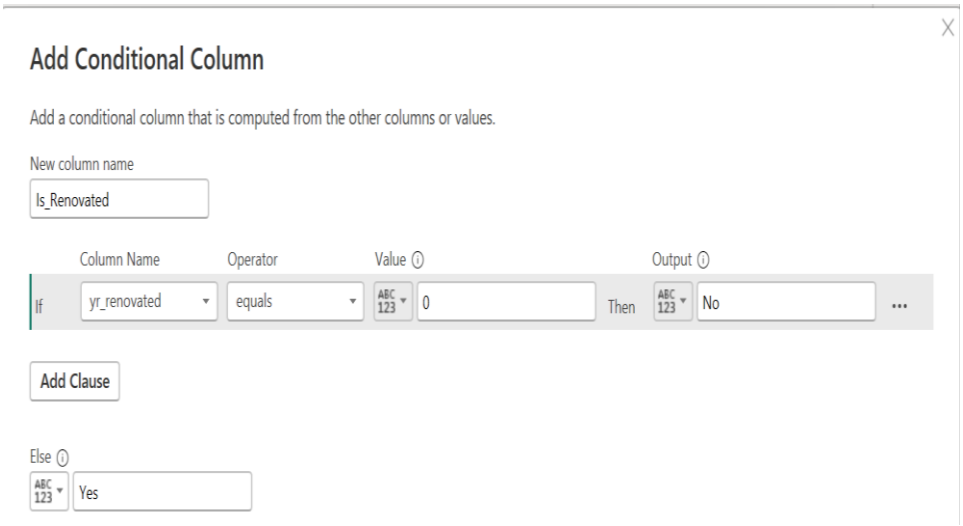
[sqft_living] + [sqft_lot]



➢ Is Renovated :

A Conditional column named Is_Renovated was created to identify whether a house had been renovated.

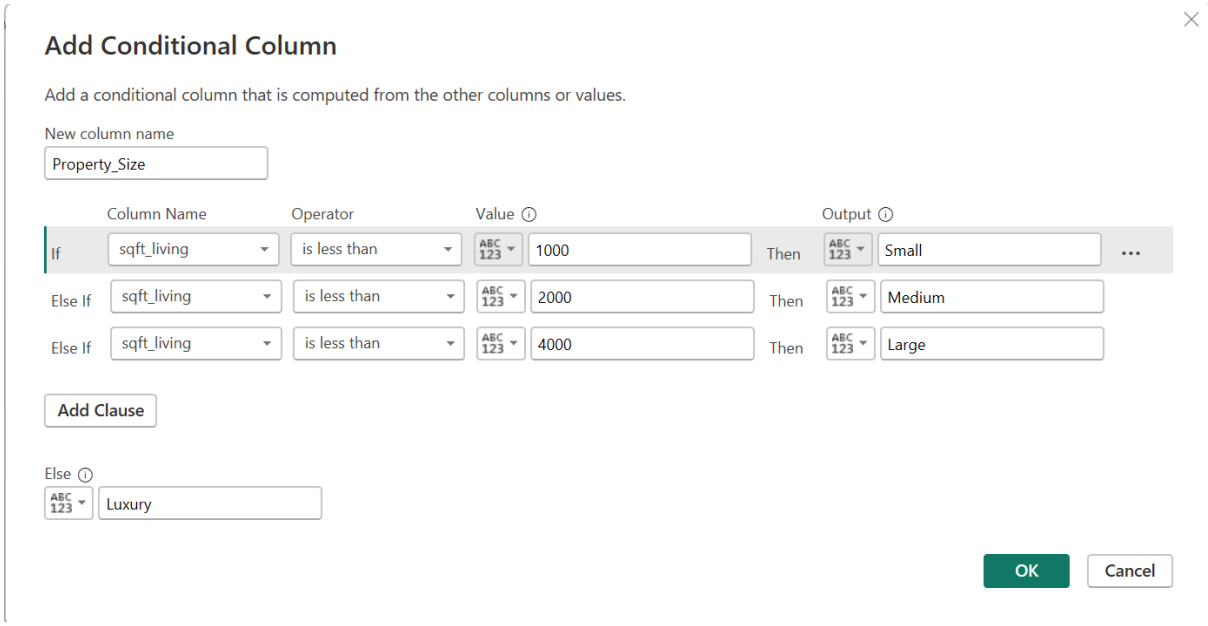
The value is "No" when the renovation year is zero and "Yes" otherwise.



➢ Property Size :

A conditional column named Property_Size was created to categorize houses according to their living area.

- **Small:** Below 1,000 square feet
- **Medium:** From 1,000 to below 2,000 square feet
- **Large:** From 2,000 to below 4,000 square feet
- **Luxury:** 4,000 square feet or more

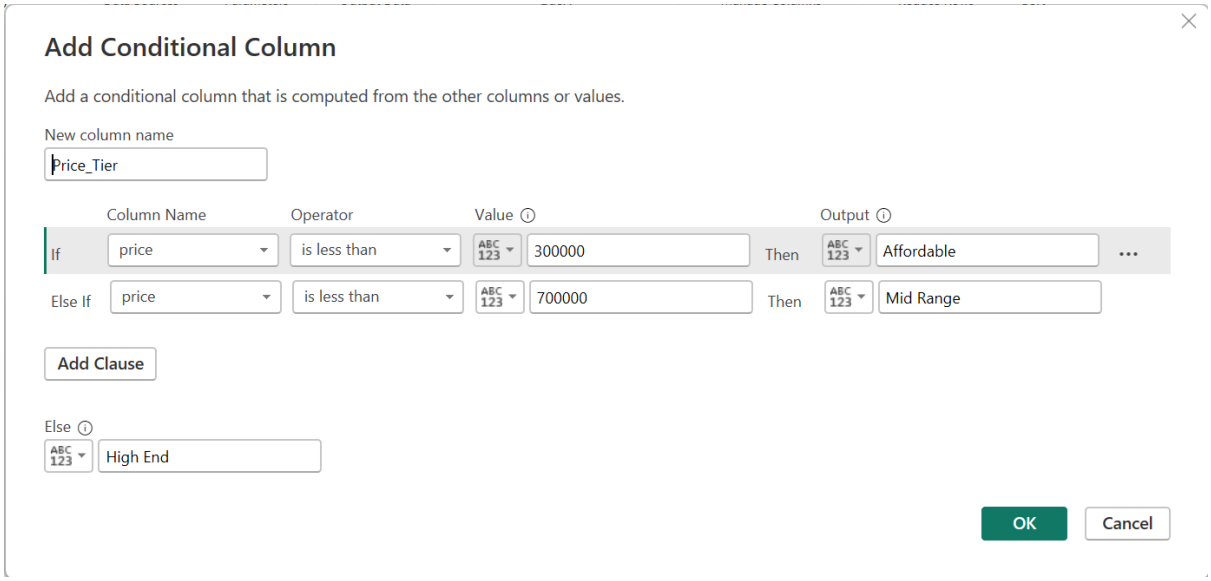


➢ Price Tier :

A conditional column named Price_Tier was created to classify houses according to their prices.

- **Affordable:** Below $300,000
- **Mid-Range:** From $300,000 to below $700,000
- **High-End:** $700,000 or more

These categories help compare houses across different price ranges.



Loading the Data :

After completing the transformations, the data was loaded into Power BI Desktop using the **Close & Apply** option. The transformed dataset was stored in a table named kc_house_data.

The table contains the original property information and the newly created columns.

Result :

The King County House Sales Dataset was imported and transformed using Power BI and Power Query Editor. New columns were created for Sale Year, Sale Month, Property Age, Price per SqFt, Total Area, Is Renovated, Property Size, and Price Tier.

Conclusion :

This project helped to understand data cleaning and transformation using Power BI. The dataset was organized, and new columns were created to make housing information easier to analyze. The transformed data is ready for further analysis and visualization.